

AERIS Expert Review Packet

Anomaly

Source / metric	openaq / Ozone (ozone)
Observed / expected baseline	0.067 / 0.0186815 ppm
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-08-02 18:00 UTC
Location	29.6239, -95.4742
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	8 / 779	20.95 to 100; mean 68.8698	34.21 at 2026-08-02 17:55Z; 5 min before event
asos	Air temperature (temperature)	degC	8 / 779	23.8889 to 37; mean 30.3363	34 at 2026-08-02 17:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	8 / 1026	0 to 360; mean 180.984	320 at 2026-08-02 17:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	8 / 1061	0 to 33.4389; mean 2.9242	2.57222 at 2026-08-02 17:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	11 / 143	5875.65 to 5947.25; mean 5902.29	5888.01 at 2026-08-02 18:00Z; at event time
noaa_gfs	Planetary boundary layer height (pbl_height)	m	11 / 143	31.9203 to 2539.72; mean 835.619	2407.74 at 2026-08-02 18:00Z; at event time
noaa_gfs	Precipitable water (precipitable_water)	mm	11 / 143	26.7445 to 57.7464; mean 39.4324	34.901 at 2026-08-02 18:00Z; at event time
noaa_gfs	Surface pressure (surface_pressure)	hPa	11 / 143	1001.37 to 1013.29; mean 1008.24	1008.5 at 2026-08-02 18:00Z; at event time
noaa_gfs	850-hPa temperature (t_850)	degC	11 / 143	18.4016 to 22.5646; mean 20.3009	19.9148 at 2026-08-02 18:00Z; at event time
noaa_gfs	10-m eastward wind (u_10m)	m/s	11 / 143	-3.62395 to 4.5116; mean 0.9894	-1.15038 at 2026-08-02 18:00Z; at event time
noaa_gfs	10-m northward wind (v_10m)	m/s	11 / 143	-3.70918 to 6.32564; mean 1.1082	-2.76918 at 2026-08-02 18:00Z; at event time
openaq	Ozone (ozone)	ppm	16 / 1039	0 to 0.081; mean 0.029	0.067 at 2026-08-02 18:00Z; at event time
openaq	PM10 (pm10)	ug/m3	3 / 217	11 to 153; mean 28.4009	14 at 2026-08-02 18:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	PM2.5 (pm25)	ug/m3	71 / 2786	0 to 50.8; mean 8.1906	7.7 at 2026-08-02 18:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	5 / 360	0 to 100; mean 27.025	96 at 2026-08-02 18:01Z; 2 min after event
openweather	Relative humidity (humidity)	percent	5 / 360	34 to 96; mean 70.0806	47 at 2026-08-02 18:01Z; 2 min after event
openweather	Precipitation rate (precipitation)	mm/h	5 / 360	0 to 0.22; mean 0.0009	0 at 2026-08-02 18:01Z; 2 min after event
openweather	Surface pressure (pressure)	hPa	5 / 360	1007 to 1014; mean 1010.8	1011 at 2026-08-02 18:01Z; 2 min after event
openweather	Air temperature (temperature)	degC	5 / 360	25.3 to 37.65; mean 31.1703	35.32 at 2026-08-02 18:01Z; 2 min after event
openweather	Wind direction (wind_direction)	degree	5 / 360	0 to 355; mean 182.858	343 at 2026-08-02 18:01Z; 2 min after event
openweather	Wind speed (wind_speed)	m/s	5 / 360	0.26 to 5.85; mean 2.1516	3.11 at 2026-08-02 18:01Z; 2 min after event
purpleair	PM2.5 (pm25)	ug/m3	73 / 5150	0 to 313; mean 10.5459	12.5 at 2026-08-02 18:00Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-08-02 18:38Z; 39 min after event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-08-02 18:38Z; 39 min after event
sentinel5p	CO column (s5p_co_column)	mol/m^2	8 / 8	0.0275078 to 0.0328455; mean 0.0306	0.0328455 at 2026-08-02 18:38Z; 39 min after event
sentinel5p	CO granules available (s5p_co_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-08-02 18:38Z; 39 min after event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	5 / 5	0.000197921 to 0.000318768; mean 0.0003	0.000306598 at 2026-08-02 18:38Z; 39 min after event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-08-02 18:38Z; 39 min after event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	4 / 4	4.01789e-05 to 5.22023e-05; mean 0	4.15036e-05 at 2026-08-02 19:10Z; 1 h 10 min after event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	4 / 4	1 to 1; mean 1	1 at 2026-08-02 19:10Z; 1 h 10 min after event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-08-02 18:38Z; 39 min after event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	5 / 5	-6.07222e-05 to 0.000178122; mean 0	-6.07222e-05 at 2026-08-02 18:38Z; 39 min after event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-08-02 18:38Z; 39 min after event
tceq	Carbon monoxide (co)	ppm	3 / 188	0 to 0.7; mean 0.1957	0.1 at 2026-08-02 17:00Z; 1 h before event
tceq	Nitrogen dioxide (no2)	ppb	12 / 735	-2.4 to 33.6; mean 4.5578	2 at 2026-08-02 18:00Z; at event time
tceq	Sulfur dioxide (so2)	ppb	3 / 187	-0.6 to 7.2; mean -0.0214	-0.2 at 2026-08-02 18:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	08-01 06Z 92.17, 12Z 74.32, 18Z 49.89, 08-02 00Z 75.74, 06Z 87.39, 12Z 61.93, 18Z 36.74, 08-03 00Z 70.06, 06Z 88.9, 12Z 68.67, 18Z 46.4, 08-04 00Z 75.82
asos	Air temperature (temperature)	08-01 06Z 25.84, 12Z 30.26, 18Z 34.18, 08-02 00Z 30.04, 06Z 27.87, 12Z 31.06, 18Z 35.13, 08-03 00Z 29.49, 06Z 26.25, 12Z 30.64, 18Z 34.42, 08-04 00Z 28.5
asos	Wind direction (wind_direction)	08-01 06Z 170.9, 12Z 235.2, 18Z 233.5, 08-02 00Z 211.6, 06Z 237.4, 12Z 127.4, 18Z 179.2, 08-03 00Z 164.5, 06Z 140.4, 12Z 158.8, 18Z 155.6, 08-04 00Z 160.5
asos	Wind speed (wind_speed)	08-01 06Z 2.194, 12Z 4.162, 18Z 3.35, 08-02 00Z 3.058, 06Z 2.814, 12Z 1.961, 18Z 2.919, 08-03 00Z 2.942, 06Z 1.526, 12Z 2.166, 18Z 4.485, 08-04 00Z 3.486
noaa_gfs	500-hPa geopotential height (gh_500)	08-01 06Z 5945, 12Z 5927, 18Z 5927, 08-02 00Z 5905, 06Z 5901, 12Z 5881, 18Z 5887, 08-03 00Z 5877, 06Z 5888, 12Z 5881, 18Z 5896, 08-04 00Z 5900, 06Z 5914
noaa_gfs	Planetary boundary layer height (pbl_height)	08-01 06Z 412.6, 12Z 369.7, 18Z 1640, 08-02 00Z 947.7, 06Z 303.3, 12Z 189.3, 18Z 2135, 08-03 00Z 1031, 06Z 184, 12Z 96.21, 18Z 2243, 08-04 00Z 997.7, 06Z 314.1
noaa_gfs	Precipitable water (precipitable_water)	08-01 06Z 38.49, 12Z 42.19, 18Z 50.15, 08-02 00Z 56.45, 06Z 51.47, 12Z 40.43, 18Z 32.12, 08-03 00Z 29.75, 06Z 31.65, 12Z 32.87, 18Z 34.35, 08-04 00Z 39.01, 06Z 33.68
noaa_gfs	Surface pressure (surface_pressure)	08-01 06Z 1011, 12Z 1011, 18Z 1009, 08-02 00Z 1005, 06Z 1008, 12Z 1008, 18Z 1008, 08-03 00Z 1005, 06Z 1008, 12Z 1009, 18Z 1009, 08-04 00Z 1007, 06Z 1009
noaa_gfs	850-hPa temperature (t_850)	08-01 06Z 22.22, 12Z 21.93, 18Z 19.69, 08-02 00Z 21.63, 06Z 20.76, 12Z 20.47, 18Z 19.48, 08-03 00Z 20.81, 06Z 19.37, 12Z 18.55, 18Z 18.95, 08-04 00Z 20.01, 06Z 20.03
noaa_gfs	10-m eastward wind (u_10m)	08-01 06Z 1.555, 12Z 2.258, 18Z 3.996, 08-02 00Z 1.967, 06Z 2.941, 12Z 1.323, 18Z -0.3522, 08-03 00Z -1.175, 06Z 0.9993, 12Z 1.264, 18Z -0.6977, 08-04 00Z -1.885, 06Z 0.6699
noaa_gfs	10-m northward wind (v_10m)	08-01 06Z 3.258, 12Z 1.821, 18Z 0.6377, 08-02 00Z 0.1537, 06Z 1.568, 12Z -1.663, 18Z -2.897, 08-03 00Z -0.8308, 06Z 2.518, 12Z 0.8871, 18Z 1.824, 08-04 00Z 4.186, 06Z 2.945
openaq	Ozone (ozone)	08-01 06Z 0.01177, 12Z 0.01407, 18Z 0.03085, 08-02 00Z 0.02182, 06Z 0.009148, 12Z 0.0239, 18Z 0.05994, 08-03 00Z 0.05262, 06Z 0.02483, 12Z 0.0207, 18Z 0.05074, 08-04 00Z 0.0222, 06Z 0.016
openaq	PM10 (pm10)	08-01 06Z 16.94, 12Z 30.22, 18Z 31.83, 08-02 00Z 24.33, 06Z 22.22, 12Z 22.89, 18Z 16.61, 08-03 00Z 20.44, 06Z 20.5, 12Z 50.18, 18Z 43.35, 08-04 00Z 41.67, 06Z 38.33
openaq	PM2.5 (pm25)	08-01 06Z 3.341, 12Z 4.897, 18Z 5.655, 08-02 00Z 8.785, 06Z 9.929, 12Z 8.068, 18Z 8.749, 08-03 00Z 10.32, 06Z 9.128, 12Z 9.872, 18Z 12.05, 08-04 00Z 11.63, 06Z 14.92
openweather	Cloud cover (cloud_cover)	08-01 06Z 4.8, 12Z 13.1, 18Z 18.45, 08-02 00Z 55.87, 06Z 67.31, 12Z 85.45, 18Z 32.53, 08-03 00Z 19.73, 06Z 1.069, 12Z 0, 18Z 2.1, 08-04 00Z 23.13
openweather	Relative humidity (humidity)	08-01 06Z 87.5, 12Z 81.24, 18Z 55.77, 08-02 00Z 71.87, 06Z 85.97, 12Z 70.94, 18Z 41.77, 08-03 00Z 65.2, 06Z 83.55, 12Z 75.97, 18Z 51.1, 08-04 00Z 71.7
openweather	Precipitation rate (precipitation)	08-01 06Z 0, 12Z 0, 18Z 0.003871, 08-02 00Z 0, 06Z 0.007586, 12Z 0, 18Z 0, 08-03 00Z 0, 06Z 0, 12Z 0, 18Z 0, 08-04 00Z 0
openweather	Surface pressure (pressure)	08-01 06Z 1013, 12Z 1014, 18Z 1010, 08-02 00Z 1009, 06Z 1010, 12Z 1012, 18Z 1009, 08-03 00Z 1009, 06Z 1011, 12Z 1012, 18Z 1010, 08-04 00Z 1011
openweather	Air temperature (temperature)	08-01 06Z 26.73, 12Z 29.56, 18Z 35.64, 08-02 00Z 31.39, 06Z 28.4, 12Z 30.66, 18Z 36.36, 08-03 00Z 31.37, 06Z 27.22, 12Z 30.23, 18Z 35.93, 08-04 00Z 30.17
openweather	Wind direction (wind_direction)	08-01 06Z 154.6, 12Z 232.2, 18Z 264, 08-02 00Z 230.7, 06Z 194.6, 12Z 90.61, 18Z 166.6, 08-03 00Z 158.4, 06Z 175.3, 12Z 215.5, 18Z 151.8, 08-04 00Z 161
openweather	Wind speed (wind_speed)	08-01 06Z 2.446, 12Z 2.695, 18Z 2.235, 08-02 00Z 2.286, 06Z 2.043, 12Z 1.942, 18Z 2.036, 08-03 00Z 1.967, 06Z 1.916, 12Z 1.322, 18Z 2.397, 08-04 00Z 2.572
purpleair	PM2.5 (pm25)	08-01 06Z 4.363, 12Z 6.291, 18Z 7.201, 08-02 00Z 12.48, 06Z 12.2, 12Z 10.37, 18Z 9.796, 08-03 00Z 12.35, 06Z 12.07, 12Z 12.31, 18Z 13.9, 08-04 00Z 13.02
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	08-01 18Z 1, 08-02 18Z 1, 08-03 18Z 1

Source	Metric	Values
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	08-01 18Z 1, 08-02 18Z 1, 08-03 18Z 1
sentinel5p	CO column (s5p_co_column)	08-01 18Z 0.02751, 08-02 18Z 0.03284, 08-03 18Z 0.03095
sentinel5p	CO granules available (s5p_co_granule_available)	08-01 18Z 1, 08-02 18Z 1, 08-03 18Z 1
sentinel5p	HCHO column (s5p_hcho_column)	08-01 18Z 0.0001979, 08-02 18Z 0.0003127, 08-03 18Z 0.0002755
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	08-01 18Z 1, 08-02 18Z 1, 08-03 18Z 1
sentinel5p	NO2 column (s5p_no2_column)	08-01 18Z 4.018e-05, 08-02 18Z 4.15e-05, 08-03 18Z 4.973e-05
sentinel5p	NO2 granules available (s5p_no2_granule_available)	08-01 18Z 1, 08-02 18Z 1, 08-03 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	08-01 18Z 1, 08-02 18Z 1, 08-03 18Z 1
sentinel5p	SO2 column (s5p_so2_column)	08-01 18Z 6.519e-06, 08-02 18Z -4.61e-05, 08-03 18Z 0.0001056
sentinel5p	SO2 granules available (s5p_so2_granule_available)	08-01 18Z 1, 08-02 18Z 1, 08-03 18Z 1
tceq	Carbon monoxide (co)	08-01 06Z 0.1611, 12Z 0.2056, 18Z 0.1812, 08-02 00Z 0.22, 06Z 0.1167, 12Z 0.1333, 18Z 0.16, 08-03 00Z 0.2444, 06Z 0.2389, 12Z 0.2722, 18Z 0.25, 08-04 00Z 0.2111, 06Z 0.1333
tceq	Nitrogen dioxide (no2)	08-01 06Z 3.42, 12Z 3.552, 18Z 4.337, 08-02 00Z 6.082, 06Z 3.036, 12Z 3.271, 18Z 2.811, 08-03 00Z 4.778, 06Z 8.049, 12Z 7.837, 18Z 4.965, 08-04 00Z 2.549, 06Z 2.683
tceq	Sulfur dioxide (so2)	08-01 06Z -0.1944, 12Z -0.1167, 18Z -0.04706, 08-02 00Z -0.18, 06Z -0.07778, 12Z -0.09444, 18Z -0.02778, 08-03 00Z -0.03333, 06Z -0.2056, 12Z 0.7444, 18Z -0.07222, 08-04 00Z -0.04444, 06Z -0.2667

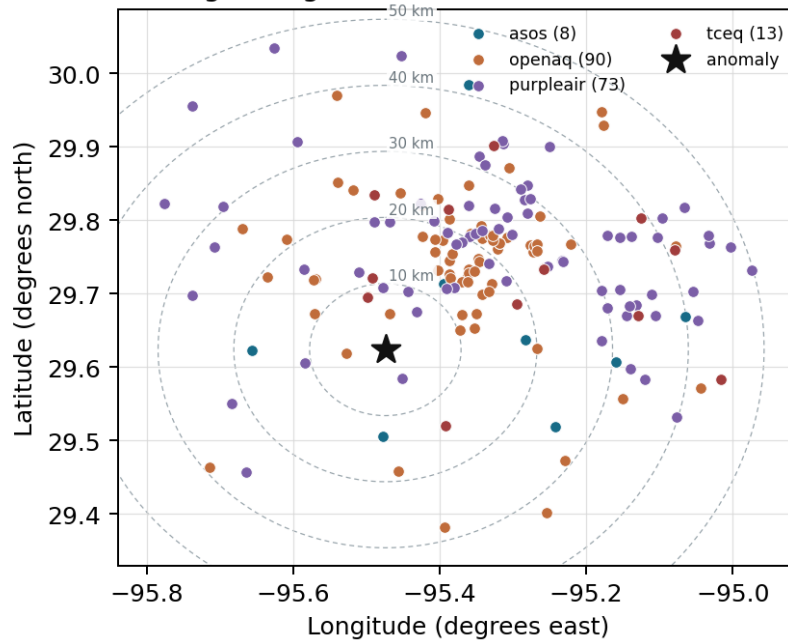
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

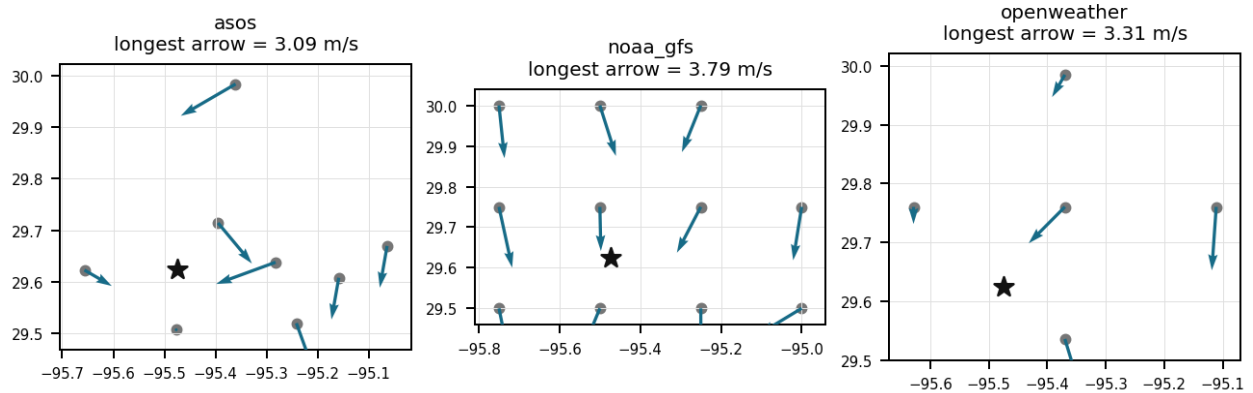
Source	Metric	Values
asos	Relative humidity (humidity)	MCJ (12.607 km) 66.34, AXH (13.101 km) 68.99, SGR (17.622 km) 73.47, HOU (18.6 km) 69.1, LVJ (25.336 km) 71.43, EFD (30.555 km) 67.37, T41 (39.95 km) 70.28, IAH (41.555 km) 64.7
asos	Air temperature (temperature)	MCJ (12.607 km) 30.58, AXH (13.101 km) 29.95, SGR (17.622 km) 29.88, HOU (18.6 km) 30.54, LVJ (25.336 km) 30.14, EFD (30.555 km) 30.84, T41 (39.95 km) 30.17, IAH (41.555 km) 30.99
asos	Wind direction (wind_direction)	MCJ (12.607 km) 204.4, AXH (13.101 km) 107.2, SGR (17.622 km) 207, HOU (18.6 km) 205.2, LVJ (25.336 km) 163.9, EFD (30.555 km) 174.5, T41 (39.95 km) 196.8, IAH (41.555 km) 190.3
asos	Wind speed (wind_speed)	MCJ (12.607 km) 3.651, AXH (13.101 km) 1.133, SGR (17.622 km) 2.897, HOU (18.6 km) 3.569, LVJ (25.336 km) 2.19, EFD (30.555 km) 4.29, T41 (39.95 km) 3.454, IAH (41.555 km) 3.04
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:29.50,-95.50 (14.001 km) 5903, gfs:29.75,-95.50 (14.241 km) 5902, gfs:29.50,-95.25 (25.691 km) 5902, gfs:29.75,-95.25 (25.8 km) 5902, gfs:29.50,-95.75 (30.023 km) 5903, gfs:29.75,-95.75 (30.107 km) 5903, gfs:30.00,-95.50 (41.894 km) 5902, gfs:30.00,-95.25 (47.083 km) 5902, +3 more
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:29.50,-95.50 (14.001 km) 760, gfs:29.75,-95.50 (14.241 km) 1085, gfs:29.50,-95.25 (25.691 km) 703.5, gfs:29.75,-95.25 (25.8 km) 1009, gfs:29.50,-95.75 (30.023 km) 752.8, gfs:29.75,-95.75 (30.107 km) 653.2, gfs:30.00,-95.50 (41.894 km) 1117, gfs:30.00,-95.25 (47.083 km) 958.6, +3 more
noaa_gfs	Precipitable water (precipitable_water)	gfs:29.50,-95.50 (14.001 km) 40.18, gfs:29.75,-95.50 (14.241 km) 39.54, gfs:29.50,-95.25 (25.691 km) 40.62, gfs:29.75,-95.25 (25.8 km) 39.98, gfs:29.50,-95.75 (30.023 km) 39.79, gfs:29.75,-95.75 (30.107 km) 38.85, gfs:30.00,-95.50 (41.894 km) 38.29, gfs:30.00,-95.25 (47.083 km) 38.82, +3 more

Source	Metric	Values
noaa_gfs	Surface pressure (surface_pressure)	gfs:29.50,-95.50 (14.001 km) 1009, gfs:29.75,-95.50 (14.241 km) 1008, gfs:29.50,-95.25 (25.691 km) 1010, gfs:29.75,-95.25 (25.8 km) 1009, gfs:29.50,-95.75 (30.023 km) 1008, gfs:29.75,-95.75 (30.107 km) 1007, gfs:30.00,-95.50 (41.894 km) 1006, gfs:30.00,-95.25 (47.083 km) 1008, +3 more
noaa_gfs	850-hPa temperature (t_850)	gfs:29.50,-95.50 (14.001 km) 20.34, gfs:29.75,-95.50 (14.241 km) 20.3, gfs:29.50,-95.25 (25.691 km) 20.21, gfs:29.75,-95.25 (25.8 km) 20.16, gfs:29.50,-95.75 (30.023 km) 20.45, gfs:29.75,-95.75 (30.107 km) 20.42, gfs:30.00,-95.50 (41.894 km) 20.49, gfs:30.00,-95.25 (47.083 km) 20.25, +3 more
noaa_gfs	10-m eastward wind (u_10m)	gfs:29.50,-95.50 (14.001 km) 0.9337, gfs:29.75,-95.50 (14.241 km) 1.098, gfs:29.50,-95.25 (25.691 km) 0.9283, gfs:29.75,-95.25 (25.8 km) 0.7845, gfs:29.50,-95.75 (30.023 km) 1.384, gfs:29.75,-95.75 (30.107 km) 1.306, gfs:30.00,-95.50 (41.894 km) 1.032, gfs:30.00,-95.25 (47.083 km) 0.3429, +3 more
noaa_gfs	10-m northward wind (v_10m)	gfs:29.50,-95.50 (14.001 km) 1.345, gfs:29.75,-95.50 (14.241 km) 1.067, gfs:29.50,-95.25 (25.691 km) 1.656, gfs:29.75,-95.25 (25.8 km) 1.06, gfs:29.50,-95.75 (30.023 km) 0.9803, gfs:29.75,-95.75 (30.107 km) 0.788, gfs:30.00,-95.50 (41.894 km) 0.8473, gfs:30.00,-95.25 (47.083 km) 0.688, +3 more
openaq	Ozone (ozone)	312 (0.0 km) 0.0303, 311 (8.352 km) 0.03034, 3055 (13.969 km) 0.02859, 1319333 (18.219 km) 0.03497, 325 (18.683 km) 0.0125, 287 (19.129 km) 0.03039, 320 (20.009 km) 0.0269, 319 (23.429 km) 0.03007, +8 more
openaq	PM10 (pm10)	15852419 (22.1 km) 18.48, 13380082 (24.246 km) 41.51, 271 (33.799 km) 25.13
openaq	PM2.5 (pm25)	15399615 (5.18 km) 7.362, 8967119 (5.464 km) 9.038, 8967021 (10.251 km) 8.895, 13955816 (10.822 km) 8.504, 9416627 (11.449 km) 5.518, 15472101 (12.224 km) 5.806, 13418566 (13.238 km) 7.463, 8966939 (13.831 km) 8.489, +63 more
openweather	Cloud cover (cloud_cover)	grid:south (14.067 km) 31.94, grid:center (18.223 km) 26.07, grid:west (21.273 km) 28.04, grid:east (38.217 km) 27.28, grid:north (41.397 km) 21.79
openweather	Relative humidity (humidity)	grid:south (14.067 km) 70.89, grid:center (18.223 km) 68.6, grid:west (21.273 km) 69.46, grid:east (38.217 km) 74.01, grid:north (41.397 km) 67.44
openweather	Precipitation rate (precipitation)	grid:south (14.067 km) 0, grid:center (18.223 km) 0, grid:west (21.273 km) 0.001667, grid:east (38.217 km) 0.003056, grid:north (41.397 km) 0
openweather	Surface pressure (pressure)	grid:south (14.067 km) 1011, grid:center (18.223 km) 1011, grid:west (21.273 km) 1011, grid:east (38.217 km) 1011, grid:north (41.397 km) 1011
openweather	Air temperature (temperature)	grid:south (14.067 km) 30.95, grid:center (18.223 km) 31.65, grid:west (21.273 km) 31.14, grid:east (38.217 km) 30.86, grid:north (41.397 km) 31.26
openweather	Wind direction (wind_direction)	grid:south (14.067 km) 213.3, grid:center (18.223 km) 200, grid:west (21.273 km) 189.7, grid:east (38.217 km) 172.3, grid:north (41.397 km) 139
openweather	Wind speed (wind_speed)	grid:south (14.067 km) 2.159, grid:center (18.223 km) 2.077, grid:west (21.273 km) 1.383, grid:east (38.217 km) 3.103, grid:north (41.397 km) 2.037
purpleair	PM2.5 (pm25)	161003 (4.842 km) 5.34, 166451 (7.009 km) 10.42, 46237 (9.253 km) 0.3681, 186315 (9.433 km) 11.82, 276512 (10.831 km) 11.63, 319092 (12.274 km) 15.48, 293735 (12.3 km) 12.05, 293725 (13.021 km) 12.37, +65 more
tceq	Carbon monoxide (co)	482011052 (22.765 km) 0.2703, 482011035 (24.228 km) 0.1667, 482011039 (33.792 km) 0.1475
tceq	Nitrogen dioxide (no2)	482010055 (8.347 km) 2.61, 482011066 (11.011 km) 7.859, 480391004 (13.954 km) 1.683, 482010416 (18.679 km) 4.703, 482011052 (22.765 km) 9.068, 482010047 (23.429 km) 3.44, 482011035 (24.228 km) 7.105, 482011039 (33.792 km) 3.05, +4 more
tceq	Sulfur dioxide (so2)	482010051 (0.017 km) -0.2492, 482011035 (24.228 km) -0.2825, 482011039 (33.792 km) 0.4836

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide

Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.

Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.

Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.

Missing or thin evidence means Unsure, not Invalid.

A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.

Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 36

The PM2.5 levels exceeded the normal range in Houston, Texas on August 2nd, indicating a significant increase in particulate matter in the air.

Mark exactly one:	☐ V	☐ I	☐ U
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Note (optional):

Claim 2 of 36

The wind direction and speed are conducive to trapping pollutants in the Houston area, exacerbating the anomaly.

Mark exactly one:	☐ V	☐ I	☐ U
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Note (optional):

Claim 3 of 36

Increased volatile organic compound activity, indicated by elevated satellite formaldehyde column amounts around 0.00031 mol/m², fueled atmospheric ozone formation in the area.

Mark exactly one:

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Note (optional):

Claim 4 of 36

West-central Houston ozone at 2026-08-02 18:00 UTC is most consistent with an aged ozone plume rather than a fresh local combustion plume because ozone reached 0.067 ppm while nearby NO₂ was only 2.0 ppb, TCEQ CO was low at 0.1 ppm within the hour, and nearby SO₂ was not elevated.

Mark exactly one:

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Note (optional):

Claim 5 of 36

Sentinel-5P satellite measurements revealed elevated total column formaldehyde concentrations on August 2 relative to August 1, indicating increased volatile organic compound reactivity that promoted rapid daytime ozone production.

Mark exactly one:

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Note (optional):

Claim 6 of 36

The pollutant involved is particulate matter (PM2.5), which is elevated to 12.5 ug/m3.

Mark exactly one:

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Note (optional):

Claim 7 of 36

The anomalous pollutant was ozone measured by openaq at 29.624, -95.474 in Houston, Texas, at 2026-08-02T18:00:00+00:00.

Mark exactly one:

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Note (optional):

Claim 8 of 36

VOC-sensitive photochemistry from urban-industrial emissions likely contributed to the ozone spike, because elevated formaldehyde and carbon monoxide columns coincided with the event while nearby NO2 and SO2 were low, a pattern consistent with an aged, oxidized ozone-producing air mass rather than a fresh combustion plume.

Mark exactly one:

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Note (optional):

Claim 9 of 36

The observed ozone value at the anomalous location and time was 0.067 ppm, compared with an expected value of 0.018681528662420382 ppm, corresponding to a z-score of 5.6520258945453925 and severe anomaly classification.

Mark exactly one:	☐ V	☐ I	☐ U
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Note (optional):

Claim 10 of 36

The 6-hour mean ozone concentration from openaq for 2026-08-02 18Z was 0.05994 ppm, which was higher than the adjacent 6-hour means of 0.0239 ppm at 2026-08-02 12Z and 0.05262 ppm at 2026-08-03 00Z within the provided context window.

Mark exactly one:	☐ V	☐ I	☐ U
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Note (optional):

Claim 11 of 36

The temperature is unusually high, causing increased emissions from industrial and vehicular sources.

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Note (optional):

Claim 12 of 36

Surface ozone concentrations in Houston reached a severe peak of 0.067 ppm at 18:00 UTC on August 2, 2026.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 13 of 36

A deep and well-mixed boundary layer supports efficient afternoon ozone buildup and vertical mixing near west-central Houston at 18:00 UTC on 2026-08-02 because modeled planetary boundary-layer height reached about 2408 m near the anomaly, far above the low morning values in the same window.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 14 of 36

Hot and dry boundary-layer conditions strongly support enhanced local ozone photochemistry in west-central Houston at 18:00 UTC on 2026-08-02 because surface temperature was about 34 to 35.3 C while relative humidity dropped to about 34 to 47 percent near the anomaly, both consistent with a strongly heated afternoon environment.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 15 of 36

Satellite column measurements over Houston detected elevated formaldehyde column densities of approximately 0.00031 mol/m^2 , indicating active precursor volatile organic compound photochemistry.

Mark exactly one:

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Note (optional):

Claim 16 of 36

Houston precursor and transport conditions support VOC-rich plume processing and recirculation because satellite HCHO and CO columns were elevated on 2026-08-02 18Z and near-surface winds were weak and variable around the anomaly time.

Mark exactly one:

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Note (optional):

Claim 17 of 36

The air-quality anomaly in Houston, Texas on August 2nd was caused by a combination of factors including high temperatures, low wind speeds, and nearby industrial activities.

Mark exactly one:

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Note (optional):

Claim 18 of 36

Deep boundary layer expansion exceeding 2,000 meters enabled efficient vertical mixing and buildup of photochemically generated ozone near the surface.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 19 of 36

Weak and shifting winds together with low contemporaneous NO₂ and elevated HCHO support transport of an aged VOC-influenced ozone plume into west-central Houston, although the case is not cleanly attributable to a fresh combustion plume because nearby NO₂ was only about 2 ppb, TCEQ CO was low, and wind direction differed between nearby observations while cloud cover was reported as high at one nearby grid point.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 20 of 36

High surface temperatures exceeding 34 °C combined with low relative humidity below 37% during the afternoon of August 2 created highly favorable atmospheric conditions for photochemical ozone formation.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 21 of 36

A nearby wildfire or agricultural burn has released pollutants into the atmosphere, contributing to the anomaly.

Mark exactly one:

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Note (optional):

Claim 22 of 36

The anomaly occurred at a location approximately 14.067 km from the query point, with an elevation of 30.95 degrees Celsius and wind direction of 213.3 degrees.

Mark exactly one:

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Note (optional):

Claim 23 of 36

The air quality in Houston, Texas is anomalous due to high levels of SO₂.

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Note (optional):

Claim 24 of 36

An ozone anomaly with a concentration of 0.067 ppm was recorded by openaq on 2026-08-02 at 18:00:00 UTC at coordinates (29.624, -95.474), reflecting a z-score of 5.65 above the expected baseline of 0.0187 ppm.

Mark exactly one:

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Note (optional):

Claim 25 of 36

Weak and shifting low-level winds likely favored recirculation and local accumulation of ozone and precursors over west-central Houston on 2026-08-02, because observed winds were only about 2 to 3 m/s with directional variability and modeled boundary-layer height was very deep by 18Z, supporting inland transport and mixing of regionally produced ozone.

Mark exactly one:

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Note (optional):

Claim 26 of 36

High daytime surface temperatures reaching up to 35 °C combined with clear conditions and low relative humidity accelerated local photochemical ozone production in Houston around 18:00 UTC on August 2, 2026.

Mark exactly one:

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Note (optional):

Claim 27 of 36

Model data from gfs shows planetary boundary layer heights expanding to between 2135 meters and 2408 meters around 18:00:00 UTC on 2026-08-02 near the anomaly location.

Mark exactly one:

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Note (optional):

Claim 28 of 36

At the time of the ozone anomaly on 2026-08-02 around 18:00:00 UTC, meteorological observations from asos and openweather showed elevated surface temperatures of 34.0 degC to 35.3 degC and relative humidity dropping to between 34.2 percent and 36.7 percent near the anomaly location.

Mark exactly one:	☐ V	☐ I	☐ U
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Note (optional):

Claim 29 of 36

The air quality in Houston, Texas is anomalous due to high levels of NO2.

Mark exactly one:	☐ V	☐ I	☐ U
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Note (optional):

Claim 30 of 36

The air quality in Houston, Texas is anomalous due to high levels of PM2.5.

Mark exactly one:	☐ V	☐ I	☐ U
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Note (optional):

Claim 31 of 36

Houston meteorology on 2026-08-02 18:00 UTC strongly favored photochemical ozone formation because surface temperature was about 35 to 36 C, humidity fell to about 34 to 47 percent near the anomaly, and GFS boundary-layer height was about 2408 m at the anomaly time.

Mark exactly one:

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Note (optional):

Claim 32 of 36

The anomaly was detected on August 2nd, 2026, at approximately 18:01:58 UTC.

Mark exactly one:

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Note (optional):

Claim 33 of 36

Hot, unusually dry afternoon conditions near Houston on 2026-08-02 likely accelerated photochemical ozone production, because surface temperature was about 34 to 35.3 C while humidity near the anomaly fell to about 34 to 47 percent at the time of the ozone peak.

Mark exactly one:

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Note (optional):

Claim 34 of 36

NO₂ levels were also elevated during the anomaly, suggesting increased emissions from industrial or vehicular sources.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 35 of 36

Meteorological observations and forecasts during the anomaly period confirmed high daytime temperatures of 35°C to 36°C, relative humidity near 34% to 37%, and a deep planetary boundary layer height exceeding 2100 meters.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 36 of 36

NOAA GFS model data show planetary boundary layer heights exceeding 2,100 meters at 18:00 UTC, supporting strong convective mixing and efficient vertical distribution of ground-level ozone.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.

